

Deploy the Application Tier (Backend API)

Deploy the Backend API.

Step 1: Connect to the Private EC2 Instance

Since the EC2 instances reside in **Private Application Subnets**, cannot connect to them directly via public SSH. must bridge through a Bastion Host or utilize **AWS Systems Manager (SSM) Session Manager**. I am using **Jump server**.

Step 2: Install Python 3 and Virtual Environment Tools

Update your package repository manager and install **Python 3**, its package manager (pip), and venv.

```
sudo apt update
```

Install pip:

```
sudo apt install python3-pip -y
```

Install Python virtual environment support:

```
sudo apt update
```

```
sudo apt install python3-venv python3-full -y
```

Create a virtual environment:

```
python3 -m venv venv
```

Activate the virtual environment:

```
source venv/bin/activate
```

Install the required Python packages:

```
pip install Flask PyMySQL gunicorn
```

Install the MySQL client to connect to the Amazon RDS database:

```
sudo apt install default-mysql-client -y
```

```
mysql --version
```

Step 3: Create the Flask Application

```
sudo nano app.py (Add your Flask application code to this file.)
```

Start the application in the background:

```
python3 app.py &
```

Test the application locally:

```
curl localhost:5000
```

Output

```
{"status": "Flask is running"}
```

Output Virtual environment

```
ubuntu@ip-192-168-3-95:~$ source venv/bin/activate
(venv) ubuntu@ip-192-168-3-95:~$ ls
app.py  venv
(venv) ubuntu@ip-192-168-3-95:~$ |
```

Output of background running

```
(venv) ubuntu@ip-192-168-3-95:~$ python3 app.py &
[1] 1238
(venv) ubuntu@ip-192-168-3-95:~$ * Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.3.95:5000
Press CTRL+C to quit
192.168.3.236 - - [08/Oct/2026 07:01:31] "GET / HTTP/1.1" 200 -
```

Output of flask running

```
(venv) ubuntu@ip-192-168-3-95:~$ curl localhost:5000
127.0.0.1 - - [08/Oct/2026 07:02:27] "GET / HTTP/1.1" 200 -
{"status": "Flask is running"}
```

Also need to configure Database Credentials in **app.py** file

```
DB_CONFIG = {
    'host': 'RDS-ENDPOINT',
    'user': 'admin',
    'password': 'YourPassword123!',
    'db': 'feedbackdb',
    'cursorclass': pymysql.cursors.DictCursor
```